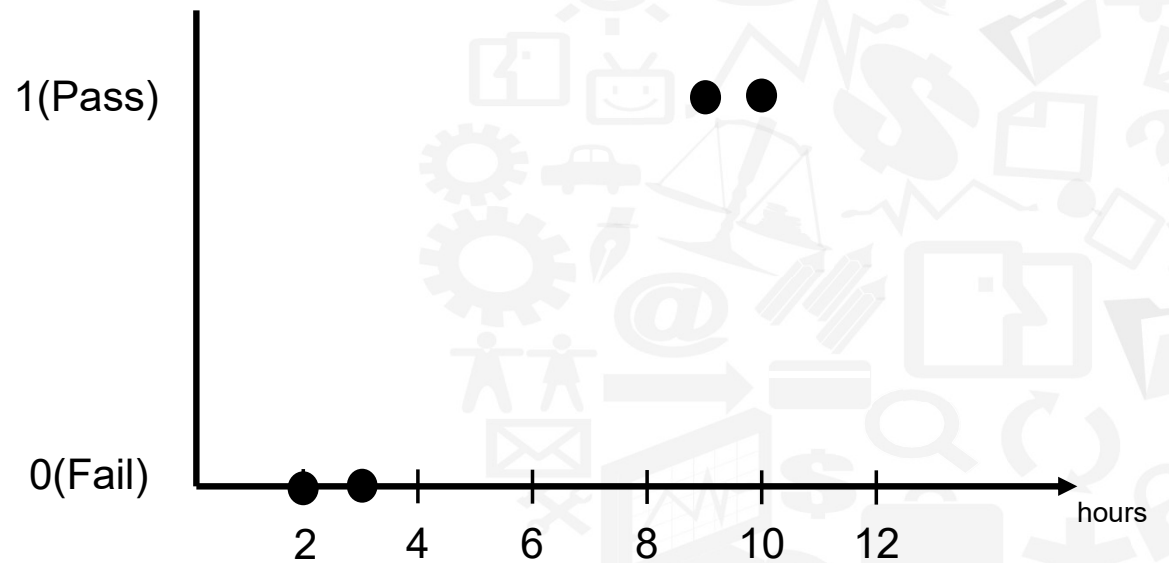




인공신경망을 이용한 이진 분류

- 학습 시간에 비례하여 합격/불합격을 구분하는 이진 분류 문제

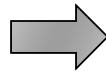
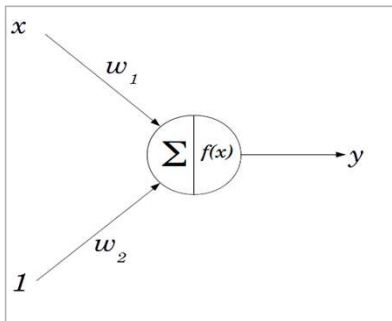
x (hours)	y (pass/fail)
10	P
9	P
3	F
2	F





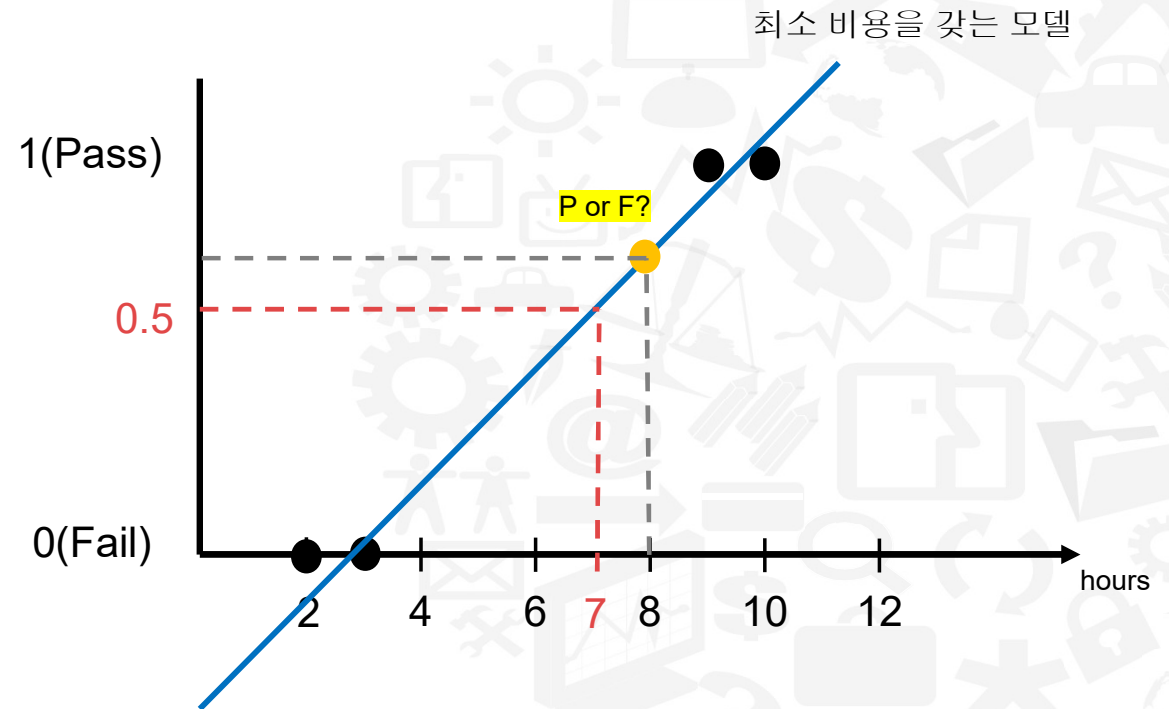
인공신경망을 이용한 이진 분류

- 회귀에서 사용한 신경망을 그대로 사용하면?



$$y = w_1x + w_2 * 1$$
$$= w_1x + b$$

w : weight or parameter
b : bias

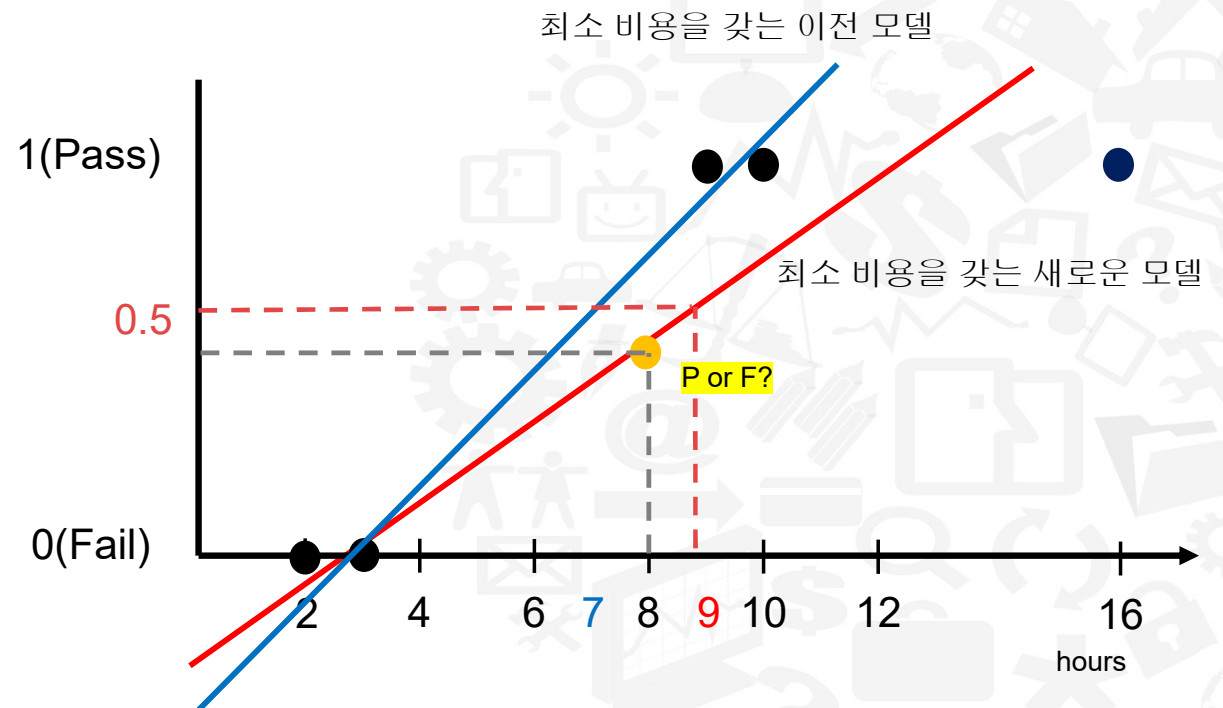




인공신경망을 이용한 이진 분류

- 학습데이터가 추가되었을 때

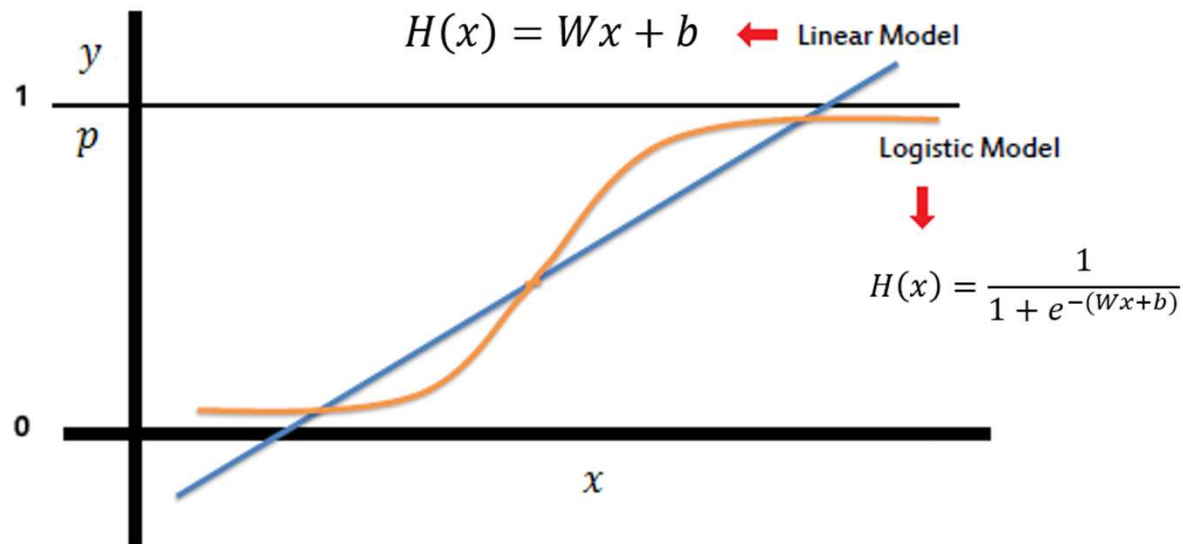
x (hours)	y (pass/fail)
10	P
9	P
3	F
2	F
16	P





인공신경망을 이용한 이진 분류

- 선형 모델에 Logistic(Sigmoid) 을 추가



Logistic function
(sigmoid function)

source: https://saedsayad.com/logistic_regression.htm



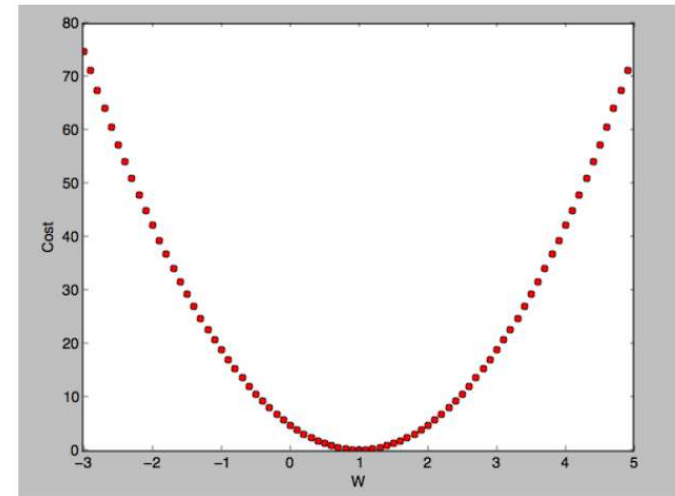
이진 분류에서 최소 비용 (손실) 찾는 법

모든 W에 대해서 계산하면...

$$\text{cost}(W) = \frac{1}{m} \sum_{i=1}^m (Wx^{(i)} - y^{(i)})^2$$

x	Y
1	1
2	2
3	3

- $W=1, \text{cost}(W)=0$
 $\frac{1}{3}((1*1-1)^2 + (1*2-2)^2 + (1*3-3)^2)$
- $W=0, \text{cost}(W)=4.67$
 $\frac{1}{3}((0*1-1)^2 + (0*2-2)^2 + (0*3-3)^2)$
- $W=2, \text{cost}(W)=?$



source: <https://hunkim.github.io/ml/>


$$H(x) = \frac{1}{1 + e^{-(Wx+b)}}$$

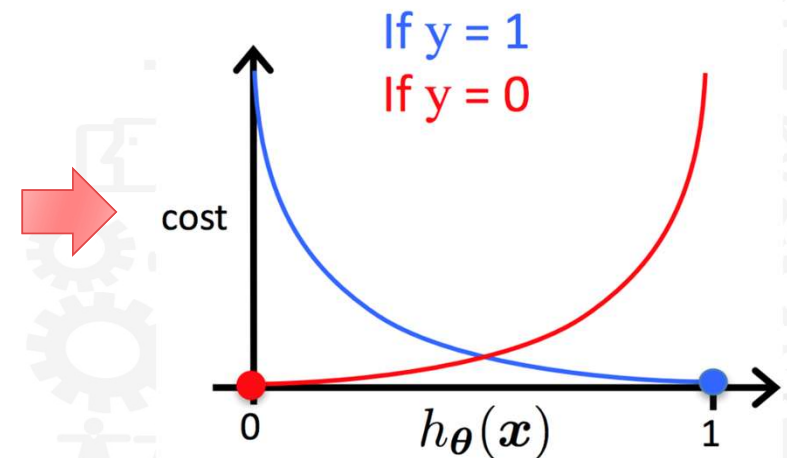
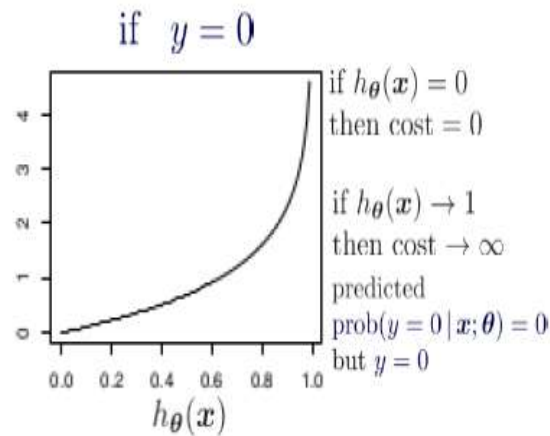
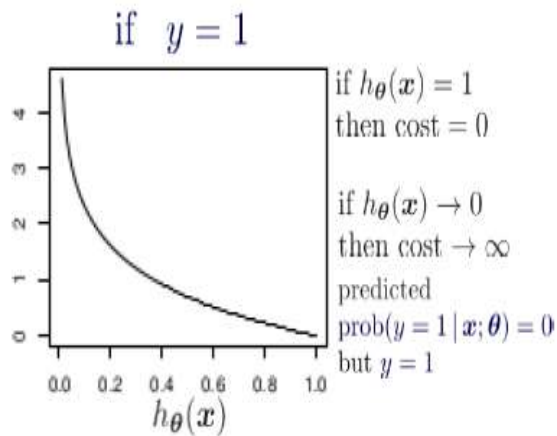
$$cost(W, b) = \frac{1}{m} \sum_{i=1}^m (H(x^{(i)}) - y^{(i)})^2$$




이진 분류에서 최소 비용 (손실) 찾는 법

로지스틱 함수를 위한 새로운 손실함수가 필요함

$$\text{cost}(h_{\theta}(x), y) = \begin{cases} -\log(h_{\theta}(x)) & \text{if } y = 1 \\ -\log(1 - h_{\theta}(x)) & \text{if } y = 0 \end{cases}$$



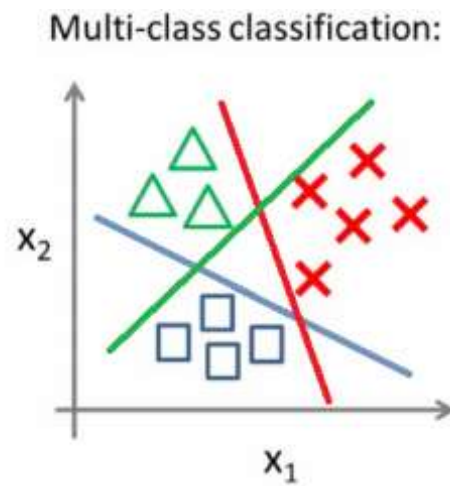
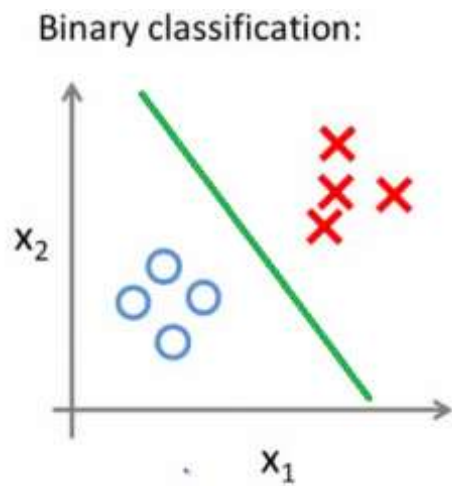
Binary Cross Entropy

$$\text{cost}(W) = \frac{1}{m} \sum (-y \log(H(x)) - (1 - y) \log(1 - H(x)))$$

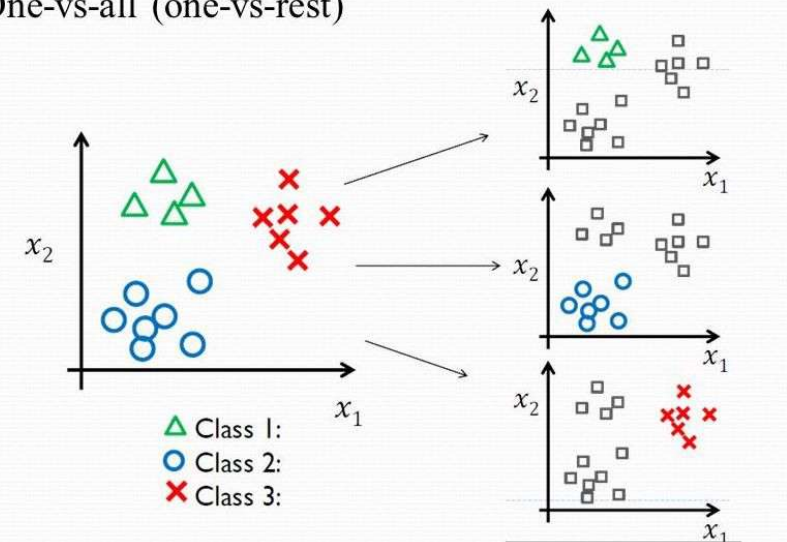


다중클래스 분류

이진 분류를 여러 개 사용하여 통합



One-vs-all (one-vs-rest)





다중클래스 분류에서의 예측

Sigmoid 함수는 2개 클래스에서 예측 확률을 출력
다중클래스 분류를 위한 다른 확률 함수가 필요



softmax function



$$\begin{bmatrix} w_1 & w_2 & w_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = [w_1x_1 + w_2x_2 + w_3x_3] = H(x)$$



$$\begin{bmatrix} w_1 & w_2 & w_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = [w_1x_1 + w_2x_2 + w_3x_3] = H(x)$$



$$\begin{bmatrix} w_1 & w_2 & w_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = [w_1x_1 + w_2x_2 + w_3x_3] = H(x)$$

XW

LOGISTIC
CLASSIFIER

$$XW = Y$$

$$\begin{bmatrix} 2.0 \\ 1.0 \\ 0.1 \end{bmatrix} \rightarrow \begin{bmatrix} 0.7 \\ 0.2 \\ 0.1 \end{bmatrix}$$
$$S(y_i) = \frac{e^{y_i}}{\sum_j e^{y_j}}$$

SCORES (logits) → PROBABILITIES



source (modified): <https://hunkim.github.io/ml/>